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$$z^2 - 7z + 13 + i = 0$$

$$D = (-7)^2 - 4 \cdot (13 + i) = -3 - 4i = (x + yi)^2 = x^2 - y^2 + 2xyi$$

$$\begin{cases} x^2 - y^2 = -3 \\ 2xy = -4 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = -3 \\ xy = -2 \end{cases} \Rightarrow \begin{cases} x^2 - y^2 = -3 \\ x^2(-y)^2 = 4 \end{cases}$$

$$\lambda^2 + 3\lambda - 4 = 0$$

$$\Delta = (3)^2 - 4 \cdot (-4) = 25 = 5^2$$

$$\lambda_1 = \frac{-3 + 5}{2} = 1$$

$$\lambda_2 = \frac{-3 - 5}{2} = -4$$

$$k = (1 + 2i)$$

$$z_1 = \frac{7 + (1 + 2i)}{2} = 4 + i$$

$$z_2 = \frac{7 - (1 + 2i)}{2} = 3 - i$$

References